

Table 1: Post-quantum KEM primitive performance on Raspberry Pi 4 (Cortex-A72 @ 1.8 GHz, 200 iterations). $\Delta P = P_{\text{mean}} - P_{\text{idle}}$ with $P_{\text{idle}} = 3309 \text{ mW}$. Sizes: public key (PK) and ciphertext (CT).

Algorithm	NIST	KeyGen (ms)	Encap (ms)	Decap (ms)	KeyGen ΔP (mW)	Encap ΔP (mW)	Decap ΔP (mW)	PK (B)	CT (B)
ML-KEM-512	L1	0.191	0.148	0.153	125.9	153.9	172.7	800	400
HQC-128	L1	22.152	44.625	73.131	412.6	441.6	566.8	2249	800
Classic McEliece 348864	L1	297.526	0.460	55.604	966.2	156.2	410.7	261120	800
ML-KEM-768	L3	0.223	0.175	0.181	149.4	185.9	205.7	1184	400
HQC-192	L3	67.430	135.390	211.278	501.2	650.9	731.5	4522	800
Classic McEliece 460896	L3	972.453	0.857	89.434	1075.7	279.6	485.1	524160	800
ML-KEM-1024	L5	0.269	0.212	0.225	154.4	180.7	177.5	1568	400
HQC-256	L5	123.836	250.537	392.216	646.9	782.9	819.0	7245	1400
Classic McEliece 8192128	L5	6901.609	1.856	209.313	1092.6	301.2	639.1	1357824	800

Table 2: Post-quantum digital signature performance on Raspberry Pi 4 (Cortex-A72 @ 1.8 GHz, 200 iterations). $\Delta P = P_{\text{mean}} - P_{\text{idle}}$ with $P_{\text{idle}} = 3309 \text{ mW}$.

Algorithm	NIST	KeyGen (ms)	Sign (ms)	Verify (ms)	KeyGen ΔP (mW)	Sign ΔP (mW)	Verify ΔP (mW)	Sig (B)
Falcon-512	L1	17.722	0.773	0.188	326.9	255.4	238.8	654
ML-DSA-44	L1	0.368	0.975	0.324	245.8	279.5	235.8	2420
SPHINCS ⁺ -128s	L1	193.609	1466.336	1.589	729.7	783.7	285.7	7856
ML-DSA-65	L3	0.551	1.212	0.476	240.9	268.9	240.2	3309
SPHINCS ⁺ -192s	L3	282.538	2610.427	2.275	797.1	763.8	303.3	16,224
Falcon-1024	L5	49.016	1.441	0.275	439.3	250.4	240.0	1267
ML-DSA-87	L5	0.739	1.626	0.718	227.0	268.1	235.2	4627
SPHINCS ⁺ -256s	L5	186.472	2323.760	3.277	715.1	790.5	255.0	29,792

Table 3: AEAD cipher performance on MAVLink packet sizes. Raspberry Pi 4, 10000 iterations, two-pass methodology (Pass 1: timing with `gc.disable()`; Pass 2: INA219 power at 100 Hz). Idle baseline: $P_{\text{idle}} = 2.961 \text{ W}$. RT $\mu\text{J} = \text{encrypt} + \text{decrypt}$ energy per round-trip operation.

Cipher	Payload (B)	Enc (μs)	Dec (μs)	Enc (MB/s)	Dec (MB/s)	Enc P (W)	Dec P (W)	Enc ($\mu\text{J}/\text{op}$)	Dec ($\mu\text{J}/\text{op}$)	RT (μJ)
AES-256-GCM	9	4.81	4.46	1.87	2.02	3.033	3.025	17.51	12.85	30.36
	28	5.13	4.89	5.45	5.72	2.958	3.325	18.14	17.49	35.63
	33	5.51	5.17	5.99	6.39	3.328	3.232	21.30	16.28	37.58
	42	5.53	5.22	7.59	8.04	2.986	2.997	19.29	15.15	34.44
	54	5.79	5.51	9.33	9.81	3.017	3.178	20.38	17.38	37.76
	263	9.80	9.59	26.83	27.41	3.078	2.983	32.81	28.65	61.45
ChaCha20-Poly1305	9	5.17	4.99	1.74	1.80	3.036	2.992	18.27	13.63	31.91
	28	5.26	4.95	5.32	5.66	3.026	3.009	17.98	14.16	32.14
	33	5.23	5.13	6.31	6.44	3.031	3.056	18.34	14.21	32.55
	42	5.19	5.00	8.10	8.40	3.049	3.249	19.04	15.42	34.46
	54	5.25	5.05	10.29	10.70	3.300	2.993	19.99	14.17	34.16
	263	6.54	6.49	40.24	40.53	3.037	3.036	22.35	18.60	40.95
Ascon-128a	9	11.36	11.16	0.79	0.81	2.970	3.040	49.13	42.41	91.53
	28	11.47	11.43	2.44	2.45	3.388	3.034	46.08	43.62	89.69
	33	11.58	11.69	2.85	2.82	3.004	2.900	49.63	42.75	92.38
	42	11.55	11.59	3.64	3.62	2.996	3.108	50.04	44.73	94.77
	54	11.69	11.79	4.62	4.58	3.363	3.041	55.62	44.44	100.06
	263	12.87	13.42	20.43	19.59	3.018	3.132	54.28	50.26	104.54

Table 4: Effect of PQC security level on DDoS detector overhead. 72 cipher suites (24 KEM×SIG pairs × 3 AEADs), aggregated per NIST level from `comparison.json` (run 2026-03-02). D-NO = no detector; D-XGB = XGBoost; D-TST = Transformer.

NIST	CPU (%)			Δ CPU (pp)		Power (mW)			Δ Power (mW)	
	D-NO	D-XGB	D-TST	+XGB	+TST	D-NO	D-XGB	D-TST	+XGB	+TST
L1 (27 suites)	21.02	46.21	89.39	25.19	68.37	3079	4004	4108	925	1029
L3 (18 suites)	22.87	48.10	89.64	25.24	66.77	3039	3996	3984	958	945
L5 (27 suites)	22.71	48.16	89.29	25.45	66.58	3084	4019	4114	936	1030
<i>All 72 suites</i>			89.44	25.29	67.24	3067	4006	4069	940	1001

Table 5: Effect of DDoS detection on PQC handshake latency. Mean handshake time (drone-side) per NIST level, averaged across all valid suites. Δ = detector overhead relative to no-detector baseline. Throughput = completed handshakes per second.

NIST	Mean Handshake (ms)			Δ Latency (ms)		Throughput (HS/s)		
	D-NO	D-XGB	D-TST	Δ_{XGB}	Δ_{TST}	D-NO	D-XGB	D-TST
L1 (27 suites)	699.37	688.86	901.32	−10.51	201.95	22.043	21.464	8.227
L3 (18 suites)	1939.10	1852.59	2791.27	−86.51	852.17	13.406	13.274	5.721
L5 (27 suites)	6582.57	4577.05	6527.46	−2005.52	−55.11	17.356	17.191	6.356
<i>All 72</i>	3215.50	2437.90	3483.60	−777.60	268.10	—	—	—

Table 6: MDEAS scheduler constants and cross-axis constraints. All values extracted from `sscheduler/policy.py` named constants.

Category	Code-defined value					
Battery thresholds	warn = 15200 mV low = 14800 mV critical = 14000 mV drain-rate warn = 500 mV/min					
Thermal thresholds	warn = 70.0 °C critical = 80.0 °C rising-rate warn = 5.0 °C/min					
Link-quality gates	min PPS = 5.0 max gap = 1000 ms max blackout count = 3					
Hysteresis timing	downgrade hold = 5.0 s upgrade cooldown = 30 s (disarmed) / 90 s (armed) AEAD recovery = 10 s					
Detector (NONE)	overhead	$\Delta P = 0.00$ W	$\Delta T = 0.0$ °C	$\Delta \text{CPU} = 0$ pp	warmup = 0.0 s	max baseline temp = 80.0 °C
Detector overhead (XG-Boost)		$\Delta P = 0.95$ W	$\Delta T = 4.8$ °C	$\Delta \text{CPU} = 35$ pp	warmup = 10.0 s	max baseline temp = 75.0 °C
Detector (TST)	overhead	$\Delta P = 1.97$ W	$\Delta T = 10.7$ °C	$\Delta \text{CPU} = 91$ pp	warmup = 5.0 s	max baseline temp = 69.0 °C
Forbidden AEAD+Detector	{Ascon-128a} with TST detector					
Discouraged KEM+Detector	{Classic McEliece 348864, 460896, 8192128} with TST detector					
Forbidden KEM+SIG pairs	All 9 Classic McEliece × SPHINCS ⁺ combinations (handshake > 2 s at baseline)					

Table 7: MDEAS graceful degradation decision boundaries. Each trigger initiates a specific axis action to reduce resource consumption while maintaining an encrypted link. Recovery proceeds in reverse with hysteresis (30/90 s) to prevent oscillation. Values from `sscheduler/policy.py`.

Trigger Condition	Axis	Action	Expected Saving	Tier
$T > 80$ °C (critical)	AEAD, DDoS	Shed TST; Ascon→ChaCha20	−2.0 W	3
$T > 75$ °C	DDoS	XGBoost → (no detector)	−0.95 W	2
$T > 72$ °C (rising)	AEAD	Preemptive AEAD downgrade	−0.05 W	1
CPU > 85%	DDoS	TST → XGBoost	−56 pp CPU	2
Battery ≤ 14800 mV	Level	L5 → L3 → L1	−7 mJ/rekey	2
Battery ≤ 14000 mV (critical)	All	Minimum config (L1 + cheapest AEAD)	Flight-hold	3
DDoS alert raised	DDoS	→ XGBoost	+951 mW	0

*Degradation tiers: 0 = Full (L5+AES-GCM+TST), 1 = Lighter AEAD, 2 = Lower level/detector, 3 = Minimum viable. Floor invariant: encrypted MAVLink link is **never** dropped to plaintext.*